Week #8 : Cell Size, Surface Area-to-Volume Ratio & Membrane Transport

Subject/الموضوع: AP Biology Unit Title: Cells Grade12: الصف

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1	I can calculate surface area, volume, and SA:V for cube-shaped cells. I can use calculated data and graphs to explain why smaller cells exchange materials more efficiently.	1.1 Flexible display of information (size, color, font, layout).	Week 8 PowerPoint — Cell Size: SA:V calculations, cell-size data challenge and graph. Review Teams / AP Biology / General channel
Lesson 2 الحصة الثانية	I can explain why volume increases faster than surface area as cells grow. I can predict how cell shape or membrane folds affect exchange efficiency.	5.3 Build fluency with graduated scaffolds and models.	Week 8 PowerPoint — Cell Size: cell adaptations, UAE heat-exchange model, AP-style questions. Review Teams / AP Biology / General channel
Lesson 3 الحصة الثالثة	I can distinguish passive and active transport using concentration-gradient direction and energy requirements. I can classify diffusion, osmosis, facilitated diffusion and membrane pumps using evidence.	6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Week 8 PowerPoint — Membrane Transport: gradients, passive versus active transport, mechanism evidence sort. Review Teams / AP Biology / General channel
Lesson 4 الحصة الرابعة	I can explain how membrane proteins and ATP help maintain concentration gradients. I can predict how reduced ATP affects active transport and ion gradients.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Week 8 PowerPoint — Membrane Transport: UAE desalination-research case, AP-style reasoning and exit ticket. Review Teams / AP Biology / General channel
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	• Bring a calculator and Biology notebook for SA:V calculations, graphs, evidence matrices and transport diagrams. • Review PowerPoint resources on Microsoft Teams. • Reinforce scientific reasoning using numerical evidence, concentration gradients and ATP requirements.		

الأسبوع# Week #8

Technology Integration دمج التكنولوجيا	• PowerPoint diagrams, cube-cell models, data tables and graphs. • Microsoft Teams for lesson resources and review; optional digital graphing.
Cross-curricular links الروابط بين المواد الدراسية	• Mathematics: surface area, volume, ratios, plotting and interpreting trends. • Chemistry: concentration gradients, molecular polarity and energy transfer. • UAE environment: heat exchange and desalination-related ion gradients.
External Assessment Integration دمج التقييمات الخارجية	AP Biology Unit 2: Cells; Topic 2.2 (Cell Structure and Function) and Topic 2.5 (Membrane Transport). AP-style calculations, model interpretation and evidence-based explanations.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Thinking: quantitative reasoning, classification, cause-effect and prediction. • Communication: labeled models, partner explanations and peer-feedback checklists. • Collaboration: paired evidence sorting and data analysis.
Standards المعايير	• Lessons 1–2: AP Biology Unit 2, Topic 2.2, LO 2.2.A; Science Practices 2.D (Models), 5.A (Calculations). • Lessons 3–4: AP Biology Unit 2, Topic 2.5, LO 2.5.A; Science Practice 3.D (Questions & Methods).
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	• Lesson 1: Why are cells generally small rather than giant? Lesson 2: How can cell shape improve exchange efficiency? Lesson 3: When can particles move without cellular energy? Lesson 4: How does ATP availability affect the ion gradients maintained by cells?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 — SA:V Calculations MUST: Calculate surface area, volume and SA:V. SHOULD: Compare ratios across cell sizes using data. COULD: Explain the effect of increasing cell size on exchange. Lesson 2 — Cell Size and Exchange MUST: State the relationship between cell size and SA:V. SHOULD: Explain how membrane folds increase exchange surface. COULD: Apply SA:V to a UAE heat-exchange model. Lesson 3 — Passive and Active Transport MUST: Identify gradient direction and distinguish passive from active transport. SHOULD: Classify mechanisms using gradient and energy evidence. COULD: Justify why protein use alone does not make transport active. Lesson 4 — Maintaining Gradients MUST: Identify ATP-dependent membrane pumps.

	SHOULD: Explain transport through channels versus pumps. COULD: Predict effects of reduced ATP on maintained ion gradients.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Respect الاحترام Heritage التراث
	المجال Area: Innovation / Sustainability / Environment
	• • Apply SA:V reasoning to heat exchange in organisms living in hot UAE environments. • Connect selective membrane transport and ion gradients to a UAE desalination-research scenario. • Use evidence-based modeling to consider how cells respond to changing external conditions.